

GLOBAL BANK CHURN ANALYSIS REPORT

Project Overview

In the banking sector, retaining existing customers is significantly more cost-effective than acquiring new ones. This project analyzes a retail banking dataset of **10,000 customers** to identify key drivers of customer attrition (churn), calculate its financial impact, and deliver an interactive Power BI dashboard. The end goal is to provide leadership with data-driven strategies to minimize revenue leakage and improve customer lifetime value (LTV).

The Business Problem

The bank is experiencing a substantial customer attrition rate. Out of 10,000 active accounts, **2,670 customers have left the bank**, resulting in a **churn rate of 26.7%**. With an overall portfolio valuation of **\$25 Million in Total Charges**, this high churn rate represents millions of dollars in lost or at-risk revenue. Management lacked clear visibility into *why* customers were leaving, *which* segments posed the highest financial risk, and *what* operational adjustments could prevent future losses.

Key Findings

- **Contract Vulnerability:** Customers on short-term **Monthly Contracts** are the dominant drivers of churn.
- **The Critical First Year:** Customer attrition peaks heavily within the **New Customer** bracket (0–12 months of tenure).
- **Payment Friction:** Customers utilizing manual or less integrated payment methods exhibit disproportionately higher churn rates than those on automated billing.
- **Financial Risk Exposure:** A substantial portion of the bank's \$25M portfolio sits in high-risk categories, requiring immediate, targeted proactive marketing interventions.

2. Business Objective

Why Customer Churn Matters

Every customer who closes their account takes away recurring monthly fees, loan potentials, and transaction volumes. Furthermore, replacing a customer costs up to **5 times more** in marketing and onboarding expenses than keeping an existing one happy.

Business Value of this Analysis

This project transforms raw transactional and demographic data into a strategic asset. By identifying exactly where the bank is losing customers, leadership can:

1. Shift from **reactive firefighting** to **proactive retention**.
2. Optimize marketing budgets by targeting only the high-risk, high-value segments.
3. Design tailored product adjustments (e.g., promotional long-term contracts) that address customer pain points before they exit.

3. Data Preparation & Engineering

To ensure data integrity, the raw dataset underwent comprehensive cleaning and feature engineering using **Power Query** within Excel and Power BI.

- **Data Cleaning & Profiling:** Handled missing data points, eliminated duplicate rows, and verified data types (e.g., ensuring financial charges were explicitly formatted as currency).
- **Feature Engineering (Calculated Columns):**
 - * **Tenure Grouping:** Segmented raw tenure numbers into clean, actionable business groups: New customer (0–12 months), Regular customer (13–36 months), and Loyal customer (37+ months).
 - **Charges Grouping:** Categorized Monthly Charges into Low, Med, and High brackets to assess if pricing structures were forcing customers away.
- **DAX Measures Implementation:** Built a robust metrics layer to power the dashboard visuals dynamically:
 - *Total Customers* = COUNT (Bank_Churn [CustomerID])
 - *Total Churn* = CALCULATE (COUNT (Bank_Churn [CustomerID]), Bank_Churn [Churn] = 1)
 - *Churn Rate* =DIVIDE ([Total Churn], [Total Customers], 0)

4. Dashboard Overview

The final product is a single-page executive-level analytical dashboard built in Power BI, optimized for intuitive navigation.

- **Executive KPI Cards:** Placed prominently at the top left to instantly give stakeholders the high-level health metrics: **Total Customers (10,000)**, **Churned Customers (2,670)**, **Churn Rate (26.7%)**, and **Total Portfolio Value (\$25M)**.
- **Interactive Global Filters (Slicers):** Enabled filtering by *Gender*, *Senior Citizen Status*, *Contract Type*, and *Risk Segment*, allowing regional managers to slice the data to match their specific operating parameters.
- **Visual Layout Strategy:** * Left side focuses on **demographics and contract breakdowns** (Who is leaving?).

- Center focus tracks **tenure and risk segments** (When and how severely are they leaving?).
- Right side focuses on **financial values and payment structures** (What is the revenue cost?).

5. Key Business Insights & Deep Dives

A. Churn by Contract Type

- **Insight:** Over 75% of all churned customers belong to the **Monthly Contract** segment. Customers locked into One-Year or Two-Year contracts show remarkably stable retention rates.
- **Management Takeaway:** Short-term flexibility is hurting our bottom line. Customers without a long-term commitment view our services as transactional rather than relational.

B. Churn by Tenure Group (The First-Year Drop-Off)

- **Insight:** Attrition is heavily concentrated among **New Customers** (those with less than 12 months with the bank). Once a customer crosses the 3-year (Loyal) mark, their propensity to churn drops to single digits.
- **Management Takeaway:** Our onboarding experience or initial expectations are failing. The first 90 to 180 days are critical; if we don't engage them deeply there, we lose them entirely.

C. Churn by Risk Segment

- **Insight:** The data engine classified a significant pool of customers under the **High Risk** segment, showing an overlap where high monthly charges correspond with declining transaction frequencies.
- **Management Takeaway:** High-paying customers are highly sensitive to fee increases or perceived poor service quality.

D. Churn by Payment Method

- **Insight:** Customers paying via manual methods (like *Mailed Checks*) churn at a significantly higher rate compared to those who set up automated billing options like *Bank Transfers* or *Credit Cards*.
- **Management Takeaway:** Manual payment options introduce a monthly "decision point" where the customer consciously reflects on whether they want to keep spending money with us. Automated billing removes this friction.

6. Strategic Recommendations

Based on the insights derived from the dashboard, the following data-informed initiatives are recommended to executive management:

1. **Introduce an Automated Billing Incentive Program:** Launch a targeted campaign offering a one-time statement credit (e.g., \$10) or small interest-rate perk to any monthly customer who switches from check/manual payments to automatic bank transfers or credit cards.
2. **Optimize the Onboarding Journey:** Task the customer experience team with building a "**First 100 Days**" Nurture Track. Introduce proactive check-in calls at Month 3 and Month 6 for all new accounts to solve friction points early.
3. **Incentivize Long-Term Contracts:** Transition existing Monthly Contract users to Annual Agreements by offering a bundled discount on monthly account maintenance fees. The upfront discount is heavily outweighed by the guaranteed retention value.
4. **Proactive Attrition Alerts for High-Risk Groups:** Establish a process where account managers receive a flag when a customer falls into the "High Risk" segment, triggering immediate outreach with personalized relationship-retention offers.

7. Realized Business Impact

By deploying this BI framework across operations, leadership gains immediate business advantages:

- **Speed to Decision:** Removes guesswork. Executives can immediately see which segments are failing month-over-month without waiting for delayed manual spreadsheets.
- **Revenue Protection:** Safekeeping even **5%** of the currently churned high-value segment translates directly into hundreds of thousands of dollars saved in pure top-line revenue.
- **Targeted Resource Allocation:** Instead of wasting marketing funds running blanket retention campaigns for all 10,000 customers, the bank can precisely direct its spending toward the vulnerable 2,670 group.

8. Core Technical Skills Demonstrated

- **Data Preparation (Excel/Power Query):** Applied robust data profiling, addressed missing values, removed system anomalies, and engineered clean business dimensions.
- **Data Modeling & DAX Engine:** Developed an efficient relational schema and wrote clean, well-commented DAX formulas to build dynamic business metrics.
- **Data Visualization & Storytelling (Power BI):** Designed an intuitive, high-contrast, uncluttered dashboard interface following corporate UI design principles.

- **Business Translation:** The ability to look at complex, segmented transactional data rows and extract clear, actionable initiatives that non-technical CEOs can confidently fund.

9. Project Conclusion & Next Steps

Summary

This project successfully bridges the gap between raw data and corporate execution. By illuminating the vulnerabilities in monthly contracts, manual payment structures, and early tenure lifecycle stages, the analysis hands the business a literal playbook for protecting its revenue portfolio.

Future Enhancements

- **Predictive Machine Learning Integration:** Integrate a Python or Azure ML model to generate a real-time probability score directly inside the dashboard, predicting exactly *which* individual customer is likely to leave next.
- **Customer Sentiment Integration:** Layer in CSAT/NPS feedback data to see how customer complaints correlate with sudden churn spikes.